Joint Angle Estimation from RGB-D Fusion for 3D Automated Ergonomic Assessment

Dwayne Lauret, Abderraouf Benali, Halim Djerroud

***Abstract*—Musculoskeletal disorders (MSDs) remain a critical concern in industrial settings. Although observational methods such as REBA and RULA are widely used for postural risk assessment, they rely heavily on expert judgment and are time-consuming. In this work, we present an automated method that combines 2D keypoint detection via OpenPose with depth information from a single RGB-D camera to estimate 3D joint angles relevant to ergonomic scoring. A polynomial regression filter and a neutral-posture calibration phase were applied to enhance robustness and reduce noise. Experimental validation on representative industrial tasks showed that 3D angle estimation yielded more consistent results with VICON reference data than the 2D approach in most of the evaluated conditions, particularly during dynamic movements involving out-of-plane rotations, a finding further corroborated by expert ergonomist annotations. Principal Component Analysis (PCA) indicated that 3D data capture a less redundancy-dominated representation of postural variability compared with 2D. Moreover, REBA 2D scores tended to produce lower risk estimates relative to REBA 3D scores in the evaluated scenarios, especially in postures involving higher-risk classifications. These findings suggest that depth-augmented 3D reconstruction may contribute to more complete postural risk detection and could be integrated into automated ergonomic assessment workflows to help prevent MSDs in real-world environments. The project webpage with demonstration videos is available at: https://hdd-robot.github.io/joint-angle-estimation/.**

***Index Terms*—3D Human Pose Estimation, Ergonomic Risk Assessment, RGB-D Sensor Fusion, Automated REBA Scoring, Musculoskeletal Disorder Prevention.**

I. Introduction

The assessment of musculoskeletal disorder (MSD) risk in the workplace often relies on Ergonomic Assessment Methods (EAMs) [1], [2], which provide structured observational frameworks for evaluating workers' postures. However, implementing a reliable tool to objectively evaluate and continuously monitor these postures remains essential for the prevention of such disorders. Such a tool could also help identify the necessary adjustments to the workspace and industrial processes to align them with workers' capabilities.

Among the most widely used EAMs are the Rapid Entire Body Assessment (REBA), the Rapid Upper Limb Assessment (RULA), and the Ovako Working Posture Analysing System (OWAS) [3]–[7]. REBA targets dynamic whole-body tasks (score 1–15, covering trunk, neck, and legs); RULA focuses on upper-limb postures in repetitive work (score 1–7); and OWAS characterizes overall posture through a four-category code. Variant methods such as MOREBA [8] and EAWS [9] extend these frameworks with additional risk factors (e.g., vibration, assembly workload), but share the same reliance on manual observation.

Within the scope of this study, REBA was selected as the primary assessment metric for three reasons. First, unlike RULA, REBA evaluates the entire body — including trunk, neck, and legs — making it more appropriate for whole-body material handling tasks. Second, REBA provides a finer-grained risk scale (1–15) compared to OWAS, which uses only four posture categories, making it more sensitive to graded posture transitions in continuous monitoring. Third, REBA's joint-angle thresholds are explicitly defined and directly mappable to measured skeletal angles, facilitating automated computation without requiring additional interpretation steps. Although these methods are well-established and widely used in industrial settings, they rely heavily on the presence of trained human observers. This reliance introduces several limitations. First, subjective judgment can lead to inter-rater variability, where different evaluators assign different scores to the same posture. Second, intra-rater and temporal inconsistencies may occur, as even the same evaluator might produce varying assessments over time. Furthermore, the presence of an observer may alter a worker's natural behavior, introducing observation bias. These constraints highlight the need for automated and objective evaluation methods capable of operating continuously in industrial environments.

A. Automatic Assessment Methods

While wearable sensors such as inertial measurement units (IMUs) [10], [11] have improved repeatability, they interfere with natural motion and remain impractical for large-scale deployment. Vision-based methods offer a non-intrusive alternative, yet according to a recent systematic review [12], approximately 67% of existing tools are still based on expert observation using methods such as REBA or RULA. Around 10% utilize structured checklists or self-reported questionnaires, while 20% leverage computer vision techniques, typically based on RGB images combined with ergonomic scoring grids. Only a small fraction (about 3%) employ direct measurement systems. These statistics underline the growing but still limited adoption of automated, vision-based systems, and motivate further research into robust 2D/3D posture analysis frameworks for ergonomic risk evaluation.

Jiao et al. [13] proposed automatic REBA assessment via multi-view RGB deep learning, combining a Cascaded Pyramid Network [14] with a temporal convolutional network [15] for 3D reconstruction, achieving 94.7% precision on industrial tasks.

Bortolini et al. [16] extended this to a network of synchronized RGB-D RealSense cameras [17], fusing multi-view

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depth maps for occlusion-resistant 3D pose estimation. As with multi-RGB approaches, deployment is hindered by the cost and complexity of multi-camera setups.

Another notable contribution is presented by Sancho-Miguel et al. [18]. They developed a real-time ergonomic risk assessment system based on 3D skeleton extraction using a single RGB-D camera. The method relies on the Microsoft Kinect v2 [19] to capture synchronized RGB and depth data, from which the subject's 3D body skeleton is extracted using the Kinect SDK. While the approach is simple, it inherits the limitations of the Kinect tracking model, particularly regarding occlusions and inaccuracies when the subject is not fully visible to the camera.

Van Crombrugge et al. [20] studied and compared three methods for automatic joint angle estimation using a VICON motion capture system as ground truth. Their approach relies on reconstructing 3D skeletons from multiple 2D or RGB-D cameras through triangulation techniques. The results demonstrated that combining triangulations from multiple camera pairs yields the highest accuracy, with an RMS error of approximately 12°, which is sufficient for reliable automated ergonomic assessment.

These recent contributions [13], [16], [18], [20]–[22] demonstrate the growing potential of automated ergonomic assessment systems based on computer vision and 3D pose estimation. However, multi-camera setups increase deployment costs, while monocular RGB-D solutions remain sensitive to occlusions and tracking artifacts. To address these challenges, we propose a method that combines 2D keypoint detection from OpenPose [23], [24] with depth sensing from a single RGB-D camera to estimate the joint angles required by the REBA assessment (neck, trunk, shoulder, arm, and knee) in 3D. Accurate alignment between depth and color images, followed by projection into real-world coordinates, enables reliable 3D skeleton reconstruction. Noise and missing data are addressed using polynomial regression filtering on sliding windows, and an initial calibration based on a neutral posture is used to reset joint angles. Overall, this real-time pipeline transforms raw sensor data into REBA scores that can be compared against those obtained by trained ergonomists, supporting objective and continuous ergonomic risk assessment in real working environments. The contributions of this work are threefold. The first is a methodological contribution: an open-source, hardware-independent single-camera RGB-D pipeline that combines OpenPose 2D keypoint detection with depth-based 3D reconstruction for multi-DoF joint angle estimation compatible with REBA and RULA scoring. Unlike proprietary SDK-based systems such as Microsoft Kinect, the framework supports finer keypoint definitions and includes a polynomial regression filter and neutral-posture calibration tailored for continuous ergonomic assessment. The second is a validation contribution: a dual protocol comparing 3D joint angles against a VICON motion capture system for angular accuracy and REBA scores against an independent certified ergonomist for score relevance, applied across a controlled single-subject trial and a 13-participant industrial data collection. The third is an analytical contribution: a Principal Component Analysis showing that 3D joint data exhibits a less projection-artifact-dominated variance structure than 2D data—preserving trunk flexion (`Torso_alpha`), a primary REBA variable absent in the leading 2D components—complemented by a correlation analysis identifying which degrees of freedom are reliably estimated from 2D and which require 3D reconstruction.

The remainder of this paper is organized as follows. Section II describes the materials and methods, including the REBA scoring methodology, the OpenPose-based 2D keypoint detection integrated with RGB-D depth data for 3D joint angle estimation, and the validation protocol against VICON motion capture data. Section III presents the experimental evaluation, including Principal Component Analysis (PCA) of 2D and 3D representations, a comparison of REBA 2D and REBA 3D scores across 13 participants, and a correlation matrix analysis. Finally, Section IV discusses the main findings, limitations, and directions for future work.

II. Materials and Methods

A. REBA Methodology

REBA (Rapid Entire Body Assessment) is a standardized method for evaluating musculoskeletal risk by combining body posture, external load, and activity characteristics into a single score. The method relies on anatomical keypoints (shoulders, elbows, wrists, hips, knees, ankles, neck and head), whose identification is traditionally performed manually by an observer. REBA splits the body into two groups. Group A covers the trunk, neck, and legs. Each segment is scored by discretizing its joint angle into predefined ranges (e.g., trunk flexion 0–20° vs. 20–60°), with added penalties for twisting or lateral bending. The three sub-scores are combined through a lookup grid into a single Group A value. Group B covers the upper arm, forearm, and wrist. Similarly, each segment is scored from its angle range, with modifiers for shoulder abduction, elevation, or wrist deviation. A second lookup grid produces a single Group B value.

The flow is as follows: Group A posture score + load adjustment = Score A; Group B posture score + coupling adjustment = Score B. Scores A and B are combined through a third grid into an intermediate value, to which the activity adjustment is added. The result is the final REBA score (typically 1–15), mapping to risk tiers from negligible (1) through very high (11+), each tied to escalating urgency of ergonomic intervention.

B. OpenPose and 3D Integration

OpenPose [23], [25], [26] is a multi-stage convolutional architecture for real-time 2D human pose estimation. Its *bottom-up* design detects keypoints independently before grouping them into full-body skeletons, making it more robust than *top-down* approaches in multi-person scenes with frequent occlusions. Built on the VGG-19 backbone [27], [28], the network produces two parallel outputs (Fig. 1): *Confidence Maps* locating body joints, and *Part Affinity Fields* (PAFs) encoding limb orientation and connectivity.

Confidence Maps $S_k(x, y)$ are 2D probability distributions for each of the 25 keypoint types in the BODY25 model

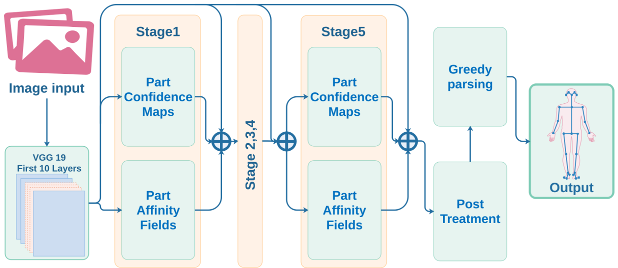


Fig. 1. Multi-stage pose estimation architecture: each stage predicts part confidence maps and part affinity fields using convolutional layers. Predictions are iteratively refined.

(Fig. 3a); local maxima are extracted via non-maximum suppression (NMS) [29] above a threshold θ. PAFs $L_c(x, y) \in \mathbb{R}^2$ encode the orientation of each limb type c[1]. For a limb connecting $\mathbf{x}_\mathrm{a}$ and $\mathbf{x}_\mathrm{b}$, the ground-truth PAF is the unit vector:

$$\mathbf{v} = \frac{\mathbf{x}_\mathrm{b} - \mathbf{x}_\mathrm{a}}{\|\mathbf{x}_\mathrm{b} - \mathbf{x}_\mathrm{a}\|_2} \tag{1}$$

This vector is assigned to all pixels within a narrow tube between $\mathbf{x}_\mathrm{a}$ and $\mathbf{x}_\mathrm{b}$, while all other pixels receive a zero vector. This generates a ground truth PAF that the network learns to approximate.

During inference, to determine whether two candidate keypoints $\mathbf{p}_a^i$ and $\mathbf{p}_b^j$ are connected, the model samples L equidistant points along the segment joining them:

$$\mathbf{p}_l = \mathbf{p}_a^i + \frac{l}{L}(\mathbf{p}_b^j - \mathbf{p}_a^i), \quad l \in \{0, \ldots, L\} \tag{2}$$

At each point $\mathbf{p}_l$, the predicted PAF vector $L_c(\mathbf{p}_l)$ is compared to the normalized direction vector of the segment. The average alignment score s_{ij} is computed as:

$$s_{ij} = \frac{1}{L} \sum_{l=0}^{L} L_c(\mathbf{p}_l) \cdot \frac{\mathbf{p}_b^j - \mathbf{p}_a^i}{\|\mathbf{p}_b^j - \mathbf{p}_a^i\|_2} \tag{3}$$

This score measures PAF–limb alignment. Coherent limb connections are selected via bipartite matching (Hungarian algorithm [30]), and a greedy parser assembles full-body skeletons, yielding anatomically consistent multi-person pose estimates.

Compared to lightweight alternatives such as MoveNet [31] and PersonLab [32] (17 keypoints, single-person), OpenPose supports up to 137 keypoints and multi-person detection at greater computational cost. As a purely 2D method, however, it cannot resolve out-of-plane rotations or sagittal-plane postures under occlusion.

To overcome this limitation, we integrated an Intel RealSense D415 depth camera, which provides synchronized depth maps $D(x, y)$ alongside RGB images. Due to the physical offset between the RGB and infrared sensors, the depth and color images must be spatially aligned. This is accomplished using the `pyrealsense2` SDK, which applies intrinsic calibration parameters to project each 2D pixel and its associated depth into real-world metric space. By combining the 2D keypoint coordinates (x, y) from OpenPose with the corresponding depth values $z = D(x, y)$, each keypoint is reconstructed in 3D space as:

$$\mathbf{P}_k^i = (x_k^i, y_k^i, z_k^i), \quad \text{with } z_k^i = D(x_k^i, y_k^i) \tag{4}$$

This results in a 3D skeleton with keypoint coordinates expressed in meters for each frame. Such multimodal fusion improves spatial fidelity in posture estimation, especially in scenarios involving sagittal-plane analysis or partial occlusions.

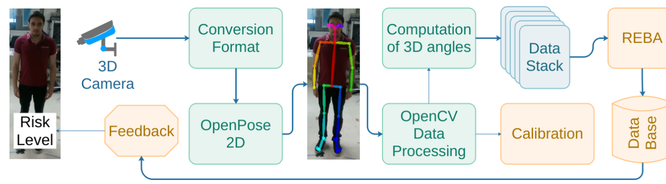


Fig. 2. Overview of the proposed pipeline. The RGB-D stream follows two parallel paths: depth-based 3D angle computation (top) and OpenPose 2D keypoint detection with OpenCV-based processing (bottom). After calibration, joint angles are aggregated and fed into the REBA scoring module, which outputs the ergonomic risk level.

For the purpose of biomechanical analysis, a symmetrical body model is assumed, and only one side (left or right) is analyzed depending on the situation. Following the REBA methodology, the analysis focuses on the following degrees of freedom: three rotational components each for the neck and trunk; two for the shoulder; and one each for the elbow and knee. Additionally, foot-ground contact is inferred to assess support conditions. To determine contact status (single or double support), we analyze the vertical (y-axis) displacement of the ankle keypoints over time. A frame is classified as "in contact" if the instantaneous ankle height is sufficiently close to the average height throughout the sequence: $|y_{current} - \bar{y}_{ankle}| < \textit{threshold}$. This thresholding strategy yields a binary contact indicator, enabling detection of postural transitions and asymmetries, which are crucial for ergonomic risk assessment.

C. Orthonormal Frames and Joint Rotations

In 3D kinematic analysis, joint orientations must be expressed in consistent orthonormal coordinate systems. In this work, we build segment-fixed geometric frames directly from OpenPose keypoints and RGB-D 3D positions.

Given a primary direction vector $\mathbf{v}_1$ (defining the main segment axis) and a secondary vector $\mathbf{v}_2$ (providing an auxiliary direction), we construct a right-handed orthonormal basis using a Gram–Schmidt-like procedure.

To estimate three-degree-of-freedom rotations (e.g., neck and trunk), we construct two local frames, denoted $\{X_A, Y_A, Z_A\}$ and $\{X_B, Y_B, Z_B\}$. Let $A = [\mathbf{X}_A\ \mathbf{Y}_A\ \mathbf{Z}_A]$ and $B = [\mathbf{X}_B\ \mathbf{Y}_B\ \mathbf{Z}_B]$ be the rotation matrices whose columns are unit axes expressed in the global coordinate system[2]. The

[1] In the BODY25 model, OpenPose defines 39 limb types, each represented by a 2D vector field (39 PAFs $\times$ 2 channels = 78 maps total), encoding the presence and orientation of each body part.

[2] $\mathbf{X}_A$ denotes the axis vector of frame A expressed in the global coordinate system.

relative rotation mapping coordinates from frame A to frame B is defined as:

$$^{A}R_B = B^\top A, \tag{5}$$

whose entries are computed by dot products:

$$^{A}R_B = \begin{bmatrix} \mathbf{X_B} \cdot \mathbf{X_A} & \mathbf{X_B} \cdot \mathbf{Y_A} & \mathbf{X_B} \cdot \mathbf{Z_A} \\ \mathbf{Y_B} \cdot \mathbf{X_A} & \mathbf{Y_B} \cdot \mathbf{Y_A} & \mathbf{Y_B} \cdot \mathbf{Z_A} \\ \mathbf{Z_B} \cdot \mathbf{X_A} & \mathbf{Z_B} \cdot \mathbf{Y_A} & \mathbf{Z_B} \cdot \mathbf{Z_A} \end{bmatrix}. \tag{6}$$

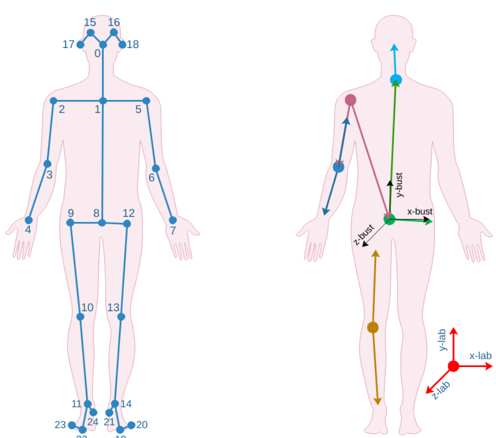
We parameterize $^{A}R_B$ using a Z–Y–X nautical convention and extract angles directly from the resulting 3×3 rotation matrix:

$$^{A}Rot_B = Rot(z_A, \gamma) \cdot Rot(y_A, \beta) \cdot Rot(x_A, \alpha)$$

Following matrix multiplication and decomposition, the nautical angles are extracted from the rotation matrix using:

$$\begin{aligned} \beta &= -\arcsin(\mathbf{Z_B} \cdot \mathbf{X_A}), \\ \alpha &= \arctan 2\left(\frac{^{A}R_B(3,2)}{\cos\beta}, \frac{^{A}R_B(3,3)}{\cos\beta}\right), \\ \gamma &= \arctan 2\left(\frac{^{A}R_B(2,1)}{\cos\beta}, \frac{^{A}R_B(1,1)}{\cos\beta}\right). \end{aligned}$$

Practical examples include the trunk frame $\{X_b, Y_b, Z_b\}$ with $Z_b \propto$ [3] $\texttt{Neck} - \texttt{MidHip}$ and $X_b \propto \texttt{RShoulder} - \texttt{LShoulder}$, and the head frame $\{X_h, Y_h, Z_h\}$ with $Z_h \propto \texttt{Neck} - \texttt{Nose}$ and $X_h \propto \texttt{REye} - \texttt{LEye}$. Frames for shoulder and wrist segments are constructed similarly by selecting appropriate primary and secondary keypoint vectors.



(a) (b)

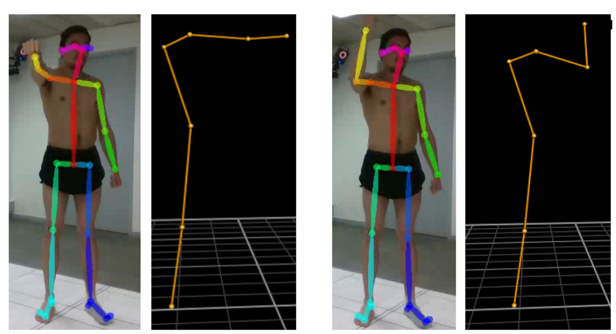
Fig. 3. (a) OpenPose BODY25 keypoint model. (b) Segment-fixed and laboratory reference frames constructed from the keypoints.

[3]The symbol $\propto$ means "is proportional to". In practice, we form the direction vector and normalize it to obtain a unit axis.

a) Keypoints Filtering and Confidence Score: To enhance measurement reliability, analysis is restricted to frames and views with sufficient keypoint detection confidence. For instance, in the right-side sagittal view, the presence of the following keypoints is mandatory: `Nose`, `Neck`, `RShoulder`, `RElbow`, `RWrist`, and `RHip`. Confidence-based filtering ensures that frames with unreliable estimates are excluded. Empirical testing suggests that a temporal smoothing window of 60 frames (equivalent to 2 seconds at 30 FPS) balances responsiveness and noise reduction effectively.

b) Calibration: Calibration is performed using the REBA-defined neutral posture, characterized by standing upright with arms relaxed at the sides. Measured joint angles in this pose are used to compute offsets for dynamic trials: $|\theta_{\text{dyn}} - \theta_{\text{calib}}| = \theta_{\text{REBA}}$. Recalibration is necessary when changing environments or repositioning the camera to maintain measurement consistency.

c) Validation Protocol: To assess joint angle accuracy, the outputs of our system were compared against two independent reference sources: (i) ground-truth kinematic data recorded using a 9-camera VICON motion capture system operating at 100 Hz, and (ii) joint angle annotations from a certified ergonomist, enabling verification that estimated angles produce consistent ergonomic risk categorization. The VICON validation was conducted with one participant. This constitutes a proof-of-concept calibration of the reconstruction pipeline rather than a statistical characterization of accuracy across individuals. Three trials of approximately three minutes each were recorded per participant. Temporal synchronization between the VICON and RealSense streams was achieved by removing the temporal offset between the two systems and resampling the VICON trajectories to match the RGB-D timestamps via linear interpolation. A known neutral posture was first used to verify the calibration offsets. Reported angular values in Table I correspond to the peak displacement reached across trials.



Frame 1 Frame 2

Fig. 4. Two representative examples of pose estimation: OpenPose overlay (left) and VICON reconstruction (right).

d) Verification of Angles: Subsequently, we quantitatively evaluated joint-angle accuracy by comparing the 3D and 2D outputs of our RealSense–OpenPose pipeline against VICON measurements during dynamic motions (Fig. 4). The comparison reported in Table I indicates that 3D angle estimates are generally more consistent with motion-capture kinematics than 2D estimates, particularly for movements where depth information helps resolve out-of-plane rotations. In practical terms, this tends to produce angles with lower frame-to-frame variability for segments that strongly influence ergonomic interpretation (e.g., neck and trunk flexion, and shoulder elevation), though the degree of improvement varies across joints as shown in Table I. Entries marked as (—) in the 2D column correspond to angles that cannot be reliably measured in 2D under our configuration, notably due to out-of-plane motion, and were therefore excluded rather than forcing potentially misleading values. Despite the overall improvement with 3D reconstruction, lateral trunk bending exhibits the largest discrepancies in both 2D and 3D, suggesting that this movement remains a limitation of the current setup, likely driven by viewpoint dependence, camera placement, and self-occlusion. Neck flexion also exhibits a substantial error in 3D mode (21.4°), which warrants specific attention given its role as a primary input to the REBA Group A sub-score. This error likely arises from two compounding factors: the neck segment is short (Neck–Nose keypoints), so small pixel-level displacements of the nose keypoint produce disproportionate rotational errors after frame decomposition; furthermore, depth measurements at the head region are susceptible to surface reflectance noise from hair and clothing. A 21° error in neck flexion is large enough to cause misclassification between REBA neck score 1 (0–20°) and score 2 ($> 20°$), and should be regarded as a current limitation of the system for neck-dominated posture scenarios. Overall, these results indicate that the 3D mode yielded lower absolute errors relative to VICON for most of the evaluated joints in the tested conditions, while acknowledging that lateral bending and neck flexion remain the most sensitive sources of residual error.

TABLE I
COMPARISON OF ANGULAR DISPLACEMENTS: VICON MOTION CAPTURE (GROUND TRUTH) VS. THE PROPOSED REALSENSE-OPENPOSE BASED SYSTEM (2D AND 3D MODES). ALL VALUES ARE IN DEGREES (°). $|\Delta|$ DENOTES THE ABSOLUTE ERROR WITH RESPECT TO VICON.

Joint movement	VICON	3D	2D	$\lvert\Delta_{3D}\rvert$	$\lvert\Delta_{2D}\rvert$
Trunk flexion	52.1	54.4	—	2.3	—
Trunk lateral	40.9	21.5	21.7	19.4	19.2
Neck flexion	40.2	18.9	—	21.4	—
Neck lateral	7.4	8.3	1.0	1.0	6.4
Shoulder elevation	70.1	65.4	—	4.7	—
Elbow flexion	85.5	63.9	57.7	21.6	27.8
Knee at rest	174.6	175.1	175.5	0.5	0.9
MAE				**10.1**	**13.6**

III. EXPERIMENTS

To address the complexity of joint angle data and quantify the added value of 3D measurements obtained using a single camera, a multi-step analytical strategy was implemented. First, PCA was used to evaluate redundancy and dimensionality in the 2D and 3D approaches and to assess whether 3D data capture a less redundant representation of posture. Second, we visualized the principal component spaces to explore how joint variability relates to risk levels. Third, we performed a systematic comparison of REBA 2D versus REBA 3D scores, with a focus on quantifying discrepancies and characterising the direction of score differences. Finally, correlation analyses were conducted to further explore the relationships between angular variables, risk scores, and errors.

Experiments evaluated the framework through four complementary analyses: (i) PCA to assess redundancy and dimensionality in 2D vs. 3D representations; (ii) visualization of principal component spaces relative to risk levels; (iii) systematic REBA 2D vs. REBA 3D score comparison; and (iv) correlation analyses across angular variables and risk scores. A static Intel RealSense D415 was positioned $\approx$2.5 m from the operator at shoulder height in a lateral (sagittal) view. Processing ran on Ubuntu 22.04 (Intel Core i9, NVIDIA RTX 4080) on a live RGB-D stream. Over 1,253 frames, mean inference times were 70.8 ± 54.7 ms (OpenPose) and 3.1 ± 0.8 ms (angle estimation + REBA), giving a total pipeline latency of 74.0 ± 54.8 ms per frame (13.5 FPS).

These results demonstrate that the proposed approach, implemented with the available hardware, operates in real time under live acquisition conditions. For ergonomic monitoring applications, where REBA scores are computed over observation windows (blocks of 30 frames, approximately 2 s) rather than on individual frames, a rate of 13.5 FPS provides sufficient temporal resolution to track most industrial movement patterns. The associated latency of approximately 74 ms means that risk scores lag behind the actual posture by roughly five to six frames, which has limited practical consequence for continuous monitoring as opposed to real-time motion control. Should higher frame rates be required for faster task dynamics, candidate optimizations include using a lighter OpenPose backbone (e.g., MobileNet-based variants), model quantization, or reducing the inference resolution of the input stream.

In line with the REBA worksheet structure, bilateral symmetry was assumed for simplicity: joint angles on the right side were considered representative of the left. Each participant (Fig. 5) performed a randomized sequence of postures, including neutral standing, dynamic tasks involving trunk bending, arm elevation, and repetitive elbow and knee flexion. These postures were selected to represent common industrial material handling movements.

REBA scores were then computed using both 2D and 3D joint angle data and compared to reference annotations provided by a certified ergonomist, who evaluated the postures based on raw frames (without OpenPose skeleton overlays). For the eight selected frames, REBA scores derived from 3D data were generally closer to the ergonomist's evaluation in most cases. Conversely, 2D-based REBA scores tended to produce lower estimates relative to REBA 3D, particularly in dynamic and non-neutral postures (Fig. 5).

These preliminary results are consistent with the added value of depth information in the posture evaluation method introduced in Section II, based on Equation (4), for ergonomic risk assessment in the evaluated conditions. Further analyses based on a 13-participant dataset and a broader range of movement patterns are presented in the following sections to corroborate these preliminary observations.

In addition, a complementary experiment was carried out involving 13 participants performing the same manual task, consisting of handling and transporting loads from point A

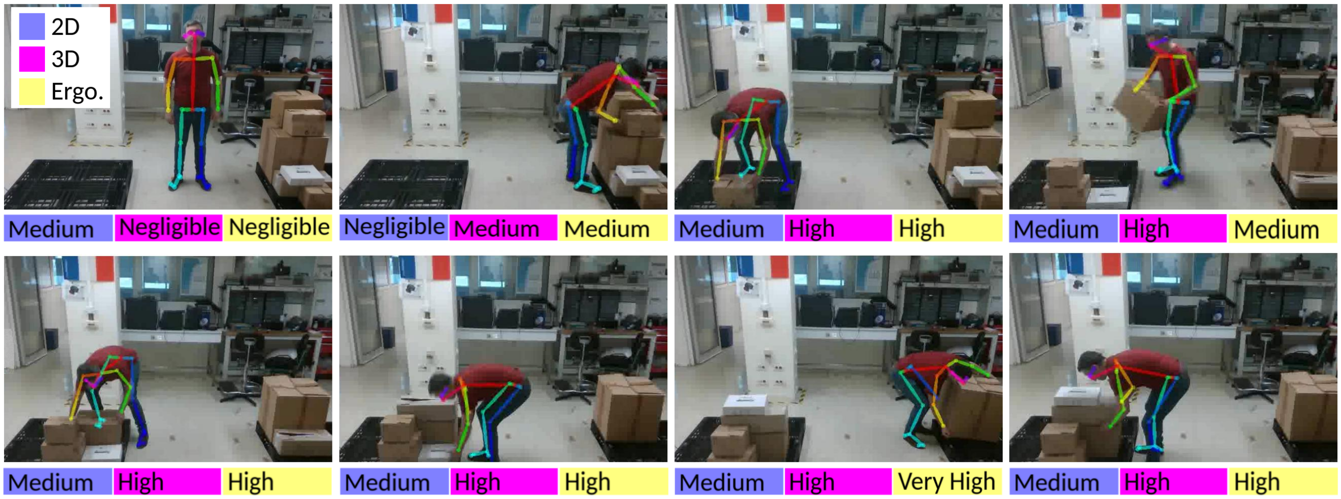


Fig. 5. Eight representative frames with automated REBA risk levels from the 2D (left label) and 3D (center label) pipelines, compared to manual scoring by an ergonomist (right label).

to point B in a randomized sequence. The participants were university students aged between 20 and 30 years, including 11 males and 2 females, all with an average body build. None reported any musculoskeletal disorders or physical impairments at the time of data collection.

This homogeneous sample provides a controlled baseline but limits generalizability. Key concerns are: (i) anthropometric variation (limb ratios, body depth) may affect keypoint localization and orthonormal frame construction; (ii) older workers or individuals with postural conditions may have different neutral-posture baselines affecting calibration; and (iii) the user-specific calibration's robustness across diverse morphologies has not been evaluated. Validation on broader industrial populations remains necessary.

A. Principal Component Analysis

The analysis aims to uncover the underlying structure of the 2D and 3D measurement approaches. Specifically, it assesses variable redundancy, determines the number of dimensions required to summarize the data, and evaluates the consistency of joint angle patterns in a reduced feature space.

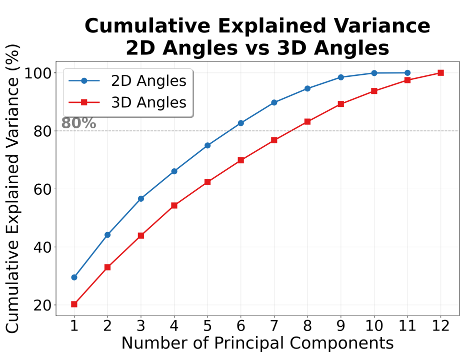


Fig. 6. Cumulative explained variance of PCA applied to 2D and 3D joint angle data. The dashed horizontal line indicates the 80 % variance threshold.

The cumulative explained variance curve (Fig. 6) shows that six principal components are sufficient to capture over 80% of the variance in 2D measurements, indicating a high degree of redundancy. In contrast, approximately eight components are required for the 3D data to reach the same threshold, suggesting a less redundancy-dominated structure in the evaluated dataset.

B. Comparison of 2D and 3D Systems via PCA Variable Loadings (Fig. 7)

The principal components (PC1–PC3) extracted from the 2D and 3D systems reveal substantial differences in how postural information is structured. In the 2D system, variance is highly concentrated in PC1 and dominated by a small set of lateral-bending variables (beta angles). In particular, `Shoulder_beta`, `Torso_beta`, and `Neck_beta` explain most of PC1, indicating that a single dominant movement pattern (apparent lateral bend) governs the representation. This concentration reflects a projection artifact where biomechanically distinct motions become artificially coupled in 2D because they are mapped onto the same image axis, causing beta variables to covary almost identically. As a result, PC2 remains largely trunk/shoulder-centric, being driven mainly by shoulder rotation and elevation (e.g., `Shoulder_gamma` and `Shoulder_elevation`), while peripheral joints such as `Left_knee` and `Left_elbow` become prominent only in PC3. Overall, this produces a compressed variance structure in which a few redundant variables dominate early components and mask limb-specific variability.

In contrast, the 3D system exhibits a more distributed loading structure in PC1, requiring a larger set of variables to explain a similar proportion of variance. Instead of being driven almost exclusively by beta angles, PC1 captures a more distributed postural combination in which `Shoulder_elevation` emerges as the leading contributor alongside `Neck_beta` and `Torso_beta`. This indicates that depth information helps separate shoulder elevation from lateral bending, revealing independent biomechanical dimensions that are conflated in 2D. A second key difference appears

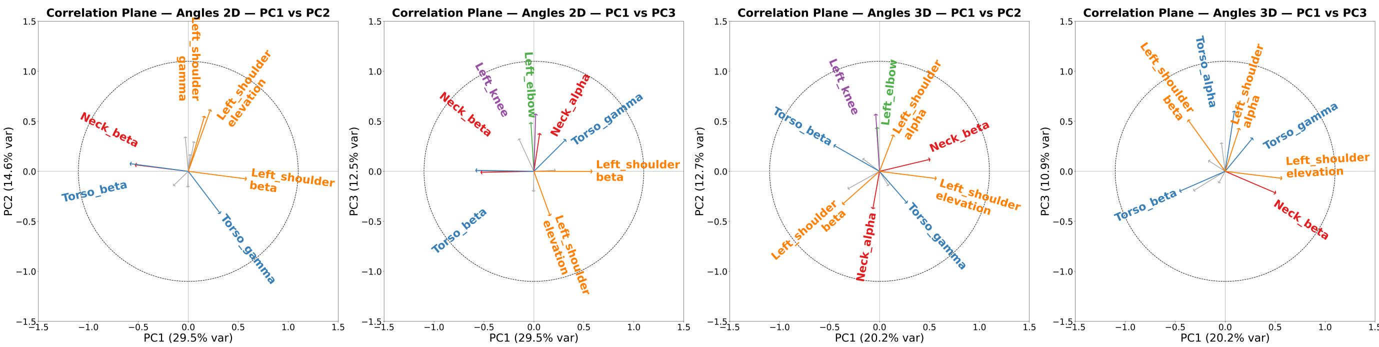


Fig. 7. PCA correlation circle (biplot) of 2D and 3D joint angle variables. Each arrow represents a variable loading; its direction and length indicate the contribution and quality of representation in the principal component plane (only loadings with $\|\mathbf{w}\| \geq 0.35$ are displayed). Arrow colors denote body segments (neck, trunk, shoulder, knee, REBA score, table height). From left to right: 2D angles in the PC1–PC2 plane, 2D angles in the PC1–PC3 plane, 3D angles in the PC1–PC2 plane, 3D angles in the PC1–PC3 plane.

in PC2: in 3D, peripheral joints (notably `Left_knee` and `Left_elbow`) contribute early, meaning their variability is represented as an independent axis rather than being deferred to residual structure. Finally, PC3 in the 3D system highlights sagittal-plane motion, with `Torso_alpha` (trunk flexion/extension) emerging as a dominant contributor, whereas it is virtually absent in the 2D PCs.

Taken together, PCA shows that 2D angles yield a representation where lateral-bend components (beta) artificially dominate and compress the variance into a few coupled variables, while 3D angles distribute variance across more independent postural dimensions. Most critically, trunk flexion (`Torso_alpha`)—a key REBA input—is captured meaningfully only in 3D mode, a structural difference that is relevant to biomechanical analysis and postural risk assessment, though it does not by itself constitute evidence of measurement accuracy.

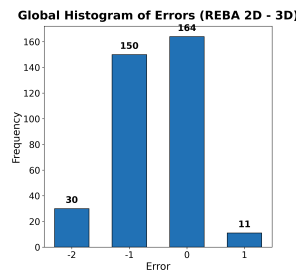


Fig. 9. Distribution of REBA score discrepancies ($\text{REBA}_{2D} - \text{REBA}_{3D}$) across all observations.

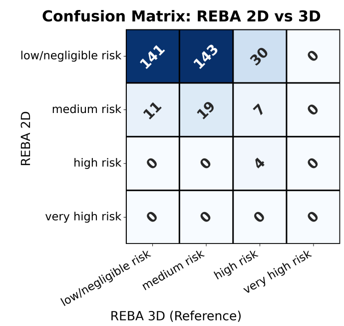


Fig. 10. Confusion matrix of REBA risk categories: 2D (predicted) vs. 3D (comparison baseline). REBA 3D serves as the engineering reference for this large-scale analysis; see the Discussion for its relationship to expert annotations.

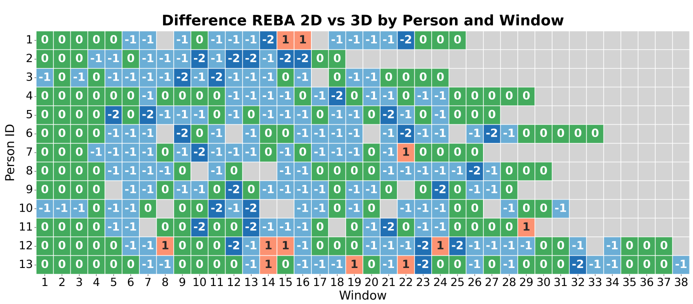


Fig. 8. Heatmap of the REBA score discrepancy ($\text{REBA}_{2D} - \text{REBA}_{3D}$) per subject and per observation window. Each cell displays the signed integer difference between the 2D and 3D estimated REBA scores; warm colors indicate overestimation by the 2D method, cool colors indicate underestimation. Missing cells correspond to windows where no valid estimate was available.

C. Analysis of Discrepancies Between REBA 2D and 3D

On the x-axis, time windows (blocks of 30 frames) are shown, over which REBA 2D and REBA 3D scores were computed (Fig. 8). The y-axis corresponds to the participant index. Each cell represents the difference between REBA 2D and REBA 3D scores for a given time window and participant.

Because participants completed the task at different paces, the number of recorded windows varies across individuals. The observed score differences range from -2 to $+1$. Overall, REBA 2D and REBA 3D often produce similar scores; however, the key question is at which risk levels this agreement occurs. If the scores align mainly in low- or no-risk postures, then the practical value of the agreement is limited.

A clear predominance of negative differences (-2 and -1) compared to positive differences ($+1$) is observed, indicating that REBA 2D tends to produce lower risk estimates relative to REBA 3D in our dataset, especially for postures labeled as high and very high risk by the 3D method. The histogram in Fig. 9 supports this observation by showing the frequency of score discrepancies across all samples. Across the 13 participants, more than 300 time windows were analyzed, and in 180 instances the 2D-based REBA score was lower than the 3D-based score.

The asymmetric distribution is consistent with a tendency of REBA 2D to produce lower risk estimates relative to REBA 3D in the evaluated dataset.

The confusion matrix in Fig. 10 quantifies the agreement between REBA 2D and REBA 3D over the 355 valid observations retained for analysis (out of 375 total windows). The matrix shows that agreement is strongest for the low/negligible-risk class: 141 low-risk windows are matched on the diagonal

(141/152 of the REBA 3D low-risk cases). In contrast, moderate and high risk are poorly recovered by the 2D scoring: most REBA 3D medium-risk cases are classified as low risk in 2D (143 instances), and a substantial portion of REBA 3D high-risk cases are also shifted downward (30 labeled low risk and 7 labeled medium risk in 2D, with only 4 correctly identified as high). This produces a pattern of lower scores for REBA 2D relative to REBA 3D, visible in the concentration of counts above the diagonal (i.e., cases where the 2D label is lower than the 3D label). No "very high risk" cases appear in either system (entire column/row equal to zero), which is consistent with the experimental conditions where REBA modifiers such as *load*, *coupling*, and *activity* were set to their minimum. Overall, 164 out of 355 samples fall on the diagonal, corresponding to an overall match rate of approximately 46%, and this limited agreement is mainly driven by the prevalence and high recall of the low-risk class.

a) Detailed Performance Analysis: Recall represents the proportion of correctly detected cases within each class according to REBA 3D. Precision is the proportion of correct predictions among those predicted as belonging to a specific class by REBA 2D. The F1-score is the harmonic mean balancing recall and precision.

TABLE II
PERFORMANCE METRICS OF REBA 2D RELATIVE TO REBA 3D.

Class	Recall (%)	Precision (%)	F1-score (%)
Low/negligible risk	92.8	44.9	60.5
Medium risk	11.7	51.4	19.1
High risk	9.8	100	17.8
Very high risk	0.0	0.0	0.0

The results (Table II) indicate a clear performance imbalance across risk categories. The *low/negligible risk* class achieves very high recall (92.8%) but only moderate precision (44.9%), yielding an F1-score of 60.5%; this suggests that most truly low-risk postures are correctly retrieved, but many samples predicted as low risk actually belong to higher-risk classes. In contrast, the *medium risk* class is poorly detected with a recall of 11.7%, despite a precision of 51.4%, resulting in a low F1-score (19.1%). The *high risk* class exhibits very low recall (9.8%), indicating substantial under-detection; however, when it is predicted, it is always correct (precision 100%), which still translates into a limited F1-score (17.8%) due to the scarcity of retrieved true positives.

D. Correlation Matrix Analysis

The analysis shown in Fig. 11 aims to evaluate the validity of 2D measurements relative to 3D data, and to explore the relationships between angular variables and overall risk (REBA) and error scores.

The analysis was structured around three objectives. First, to assess the validity of using 2D measurements as surrogates for their 3D counterparts by directly examining their correlations. Second, to better understand how joint angles relate to overall risk metrics such as REBA and error, including predictive modeling through regression. Third, to evaluate redundancy in the measurements to determine whether certain dimensions can be eliminated or if REBA scores depend excessively on a limited number of variables.

a) Bootstrap Methodology: To assess correlation robustness, the dataset (≈300 observations) was resampled 1,000 times and the full correlation matrix recomputed each time. A correlation was retained if its 95 % bootstrap confidence interval excluded zero and $|r| \geq 0.3$. Of 496 unique variable pairs, 80 met both criteria, indicating a non-trivial dependence structure.

b) Within-modality redundancy (Figs. 11a–b): Both modalities exhibit inter-variable redundancy. In 2D, Torso_beta_2D, Left_shoulder_beta_2D, and Neck_beta_2D form a near-collinear cluster, with additional couplings between trunk rotation (Torso_gamma_2D), and shoulder internal descriptors (Left_shoulder_gamma_2D, Left_shoulder_elevation_2D). In 3D the structure is less extreme but organized around shoulder elevation: Torso_beta_3D and Neck_beta_3D both correlate with Left_shoulder_elevation_3D, and a trunk flexion–rotation coupling persists. In both cases, a subset of variables co-vary and may not provide fully independent information for downstream modeling.

c) 2D–3D validity (cross-correlations) (Fig. 11c): To evaluate the validity of 2D measurements as surrogates for 3D angles, cross-correlations between 2D and 3D variables were examined. Some angles showed near-perfect agreement between modalities: Left_knee_2D vs. Left_knee_3D, Left_elbow_2D vs. Left_elbow_3D, and Left_shoulder_elevation_2D vs. Left_shoulder_elevation_3D. Beyond these direct correspondences, meaningful cross-links were also detected, indicating that certain 2D variables partially reflect 3D postural configurations. For example, Torso_beta_2D correlated with Torso_beta_3D, and Torso_gamma_2D correlated with Torso_gamma_3D. Several robust associations linked shoulder and trunk descriptors across modalities (e.g., Torso_gamma_2D vs. Left_shoulder_beta_3D; Left_shoulder_beta_2D vs. Left_shoulder_beta_3D). The presence of both (i) very high same-joint 2D–3D correlations for some joints and (ii) moderate cross-links for others suggests that 2D may be an excellent proxy for certain angles (knee, elbow, and shoulder elevation), whereas other degrees of freedom are only partially captured and may depend more strongly on viewpoint and kinematic coupling.

d) Interpretation of the correlation structure: Taken together, these results reveal three main points. First, the 2D measurements contain a strong redundancy: trunk flexion, neck flexion, and shoulder β form an almost collinear cluster, meaning that in this dataset these variables largely describe the same underlying postural variation rather than independent joint behavior. Second, the 3D angles show a similar but less extreme structure, with a coherent cluster organized around shoulder elevation, suggesting that elevation in 3D is systematically coupled with trunk and neck configurations during the recorded tasks. Third, the cross-correlations indicate that 2D can be a highly reliable surrogate for certain degrees of freedom (knee, elbow, and shoulder elevation, with near-perfect 2D–3D agreement), while other components are only moderately transferable (e.g., trunk rotation and shoulder orientation parameters). Practically, this implies that some angles

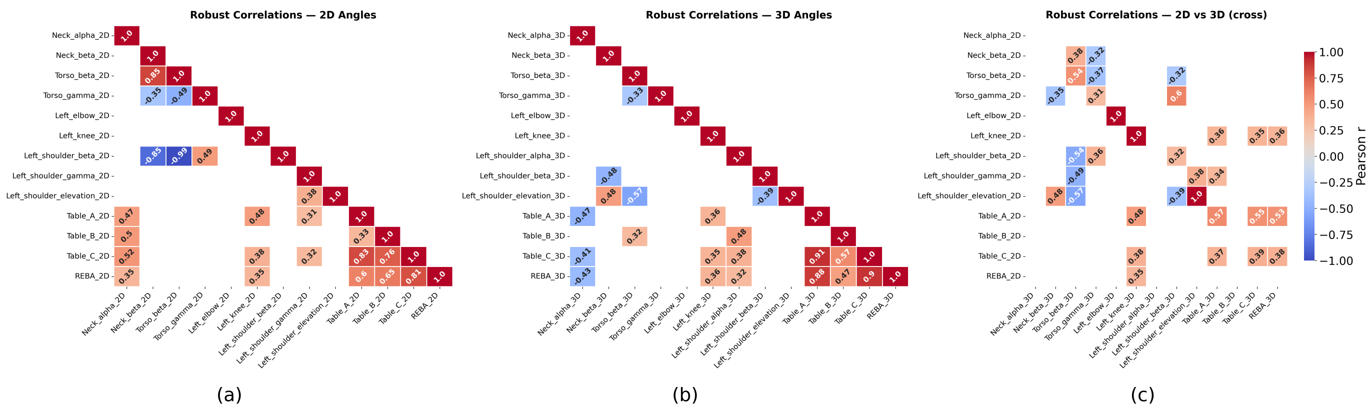


Fig. 11. Pearson correlation matrices of joint angles and REBA scores, retaining only robust correlations ($|r| > 0.3$, bootstrap 95 % confidence interval excluding zero, $N = 1000$ resamples). (a) Correlations among 2D-estimated variables. (b) Correlations among 3D-estimated variables. (c) Cross-correlations between 2D and 3D variables. Color encodes the sign and magnitude of the correlation coefficient (blue: negative, red: positive).

can be estimated confidently from 2D, whereas others may require 3D or careful camera/viewpoint control.

IV. Discussion and Conclusion

Our approach, which combines 2D detection via OpenPose with depth information from the Intel RealSense D415, provides a 3D skeletal reconstruction pipeline and demonstrates its performance in the evaluated setup. The use of polynomial regression on sliding windows helps reduce noise and mitigate the impact of missing frames. However, the system still has limitations: environmental changes may require recalibration, and certain occlusion scenarios (e.g., object handling in front of the torso) remain challenging. Although the confidence-based filtering and polynomial regression over sliding windows provide partial robustness to isolated missing frames, no systematic evaluation under controlled occlusion conditions was performed in this study. Further developments are needed to address complex scenarios such as multi-person tracking, variable lighting conditions, and partial visibility of the worker. Variable lighting can be partially mitigated using the infrared-based depth channel, which is less sensitive to illumination changes than RGB. Dynamic backgrounds may be handled through depth-based foreground segmentation, while multi-person environments would require individual tracking combined with per-skeleton REBA scoring.

Before interpreting the results, the role of each reference must be clarified. The expert ergonomist provides the closest approximation to a ground truth but annotated only eight frames, precluding use as a reference for the full 355-window confusion matrix. REBA 3D therefore serves as an engineering comparison baseline, justified by: (i) closer agreement with VICON angular estimates; (ii) closer alignment with expert annotations on the eight annotated frames; and (iii) a more distributed, less redundant postural representation confirmed by PCA and correlation analysis. This does not constitute independent validation of REBA 3D as a gold standard.

The confusion matrix (Fig. 10) shows that REBA 2D achieves high recall for low-risk postures (92.8%) but fails to detect medium and high-risk cases (recall 11.7% and 9.8%, respectively), yielding an overall match rate of ≈46% driven by class imbalance. This is consistent with the observed tendency of REBA 2D to produce lower risk estimates relative to REBA 3D, particularly at higher-risk levels.

Furthermore, the correlation matrix analysis indicates that 2D and 3D joint angles are broadly consistent, with strong cross-correlations for several corresponding angle pairs. In particular, knee, elbow, and shoulder elevation showed near-perfect agreement between modalities, suggesting that 2D can be a useful proxy for these degrees of freedom under similar viewpoints and task conditions. In contrast, other components (notably trunk rotation and some shoulder orientation descriptors) exhibited only moderate cross-correlations, suggesting partial capture by 2D and greater sensitivity to viewpoint and inter-joint coupling. The within-modality correlation structure also revealed clustered dependencies especially in the 2D set, highlighting redundancy among predictors. Overall, the associations observed in this correlation-based analysis were predominantly linear and interpretable.

A methodological limitation concerns upper-limb laterality in the REBA scoring pipeline. Group B is currently computed for a single side, following the REBA guideline to score the limb performing the primary task. In an automated setting, this implicitly assumes the selected side is representative of overall upper-limb exposure, an assumption that may not hold in asymmetric tasks where the two arms adopt distinct postures. This limitation is compounded in monocular RGB-D capture, where the more loaded limb may be more frequently self-occluded, biasing scoring toward the better-observed side. Future work should compute Group B for both limbs, propagate a per-side confidence measure derived from keypoint reliability and depth validity, and validate against bilateral VICON reference data on explicitly asymmetric tasks.

Our study presents a framework for the partial automation of the REBA method, combining 2D joint detection using OpenPose with depth-based 3D reconstruction via an Intel RealSense D415 camera. By constructing orthonormal reference frames for each body segment and extracting both single-axis and nautical angles, the system supports continuous posture evaluation in the evaluated industrial setting. Through systematic validation including comparison with VICON reference data, expert ergonomist annotations, and statistical analyses, our results indicate that the proposed 3D system yielded more consistent joint angle estimates relative to VICON and produced risk scores that aligned more closely with expert assessments than the 2D-only approach, in the scenarios considered.

PCA further shows that 3D data capture a less redundant, more dimensionally distributed representation of posture than 2D; it should be noted that this is a finding about data *structure*, not measurement accuracy — validation of accuracy remains grounded in the VICON comparison. Importantly, REBA 3D scores aligned more closely with expert evaluations on the annotated frames, and the tendency of REBA 2D to produce lower scores relative to REBA 3D was more pronounced in higher-risk postures. For context, the joint-angle MAE achieved in 3D mode (10.1°) is comparable to the $\sim$12° RMS reported by Van Crombrugge et al. [20] using multi-camera triangulation, suggesting that single RGB-D camera fusion can approach multi-camera accuracy for most joints under controlled conditions. Direct comparison of REBA classification performance with the 94.7% precision reported by Jiao et al. [13] is not straightforward, given differences in ground-truth definitions, experimental protocols, and datasets; establishing a common evaluation benchmark remains an open challenge for the field. These findings highlight the potential of 3D-enhanced ergonomic assessment to contribute to more complete identification of postural risks and to the prevention of musculoskeletal disorders (MSDs) in industrial environments.

Future work will focus on improving occlusion handling, extending the system to multi-person tracking and more dynamic task contexts, incorporating loading and coupling factors into risk evaluation, and conducting larger validation campaigns across multiple operators and work settings.

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